

Question Bank

Unit 4: Polymer

Fundamentals

1. Define polymer. How does polymerization occur?
2. Differentiate between monomer, oligomer, and polymer.
3. Explain the difference between addition polymerization and condensation polymerization with examples.
4. What are copolymers? Give two examples.
5. Distinguish between natural and synthetic polymers.

Types of Polymers

6. Classify polymers based on:
 - Source
 - Structure
 - Mode of polymerization
7. Differentiate between thermoplastic and thermosetting polymers with examples.
8. Write the characteristics of thermoplastic polymers.
9. Write the characteristics of thermosetting polymers.
10. Why are thermosetting plastics used in electrical insulations?

Industrially Important Polymers

11. Discuss the preparation and applications of natural rubber.
12. How is Buna-S prepared? Mention its applications.
13. What is Buna-N? Write its preparation and applications.
14. Write the preparation, properties, and applications of neoprene.
15. Discuss the preparation and applications of isoprene rubber.
16. How is Nylon-6 prepared? Mention two of its applications.
17. Differentiate between Nylon-6 and Nylon-6,6 in terms of preparation and properties.
18. What is Dacron (or Terylene)? How is it prepared?
19. Compare natural fibers (cotton, wool) with synthetic polymers (nylon, terylene).
20. What are the advantages of synthetic polymers in the textile industry?

Advanced Topics

21. Define conducting polymers. Give examples and applications.

22. What are biodegradable polymers? Why are they important?
23. Explain how biodegradable polymers can solve environmental issues.
24. Discuss the role of polymers in biomedical applications.
25. Write short notes on:
 - a) Bioplastics
 - b) Polymer composites